이름: 박서희
학번: 2025045332
날짜: 2026.10.23

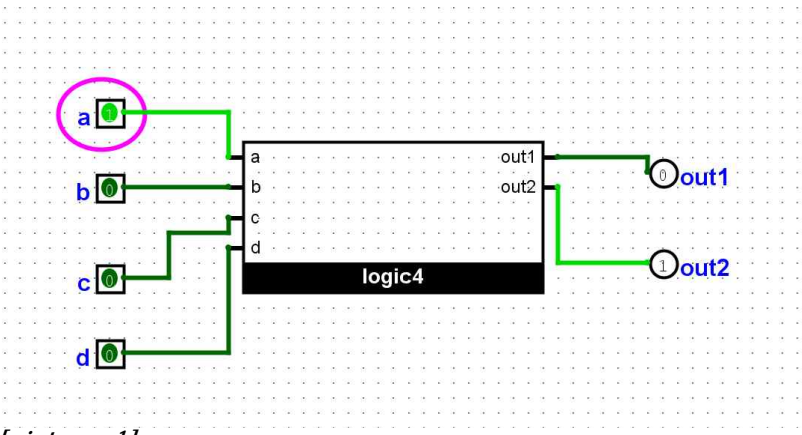
VHDL Code_1

```
 library ieee;
use ieee.std_logic_1164.all;

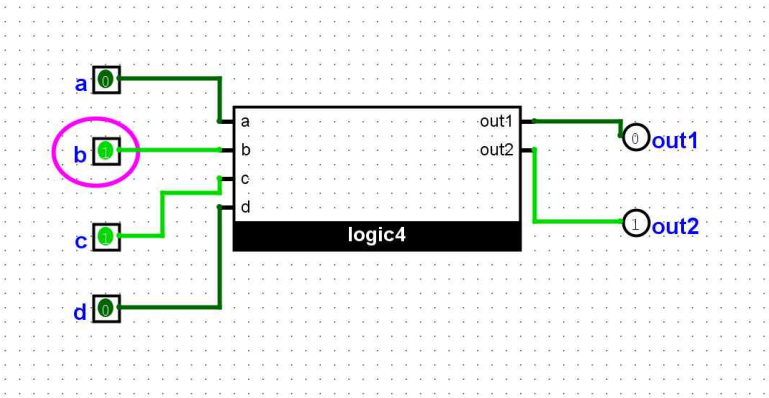
entity logic4 is
	port (
		a, b, c, d : in std_logic;
		out1, out2 : out std_logic
	);
end entity logic4;

architecture bhv of logic4 is
begin
	out1 <= a and b and c and d;
	out2 <= a or b or c or d;
end architecture bhv;
```

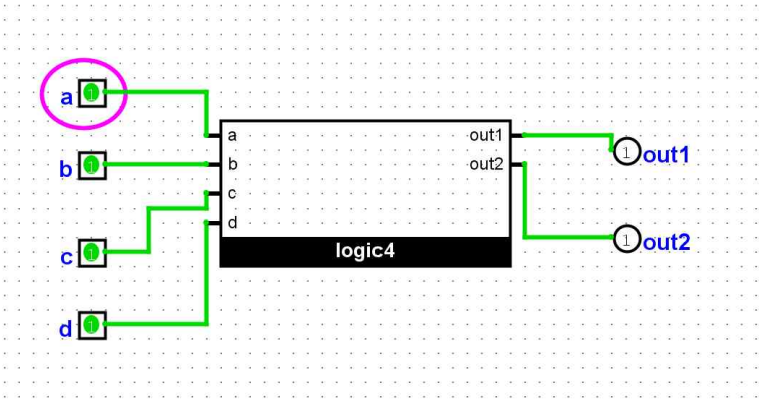
Simulation Screendshots



[picture 1]



[picture 2]



[picture 3]

VHDL Code_2

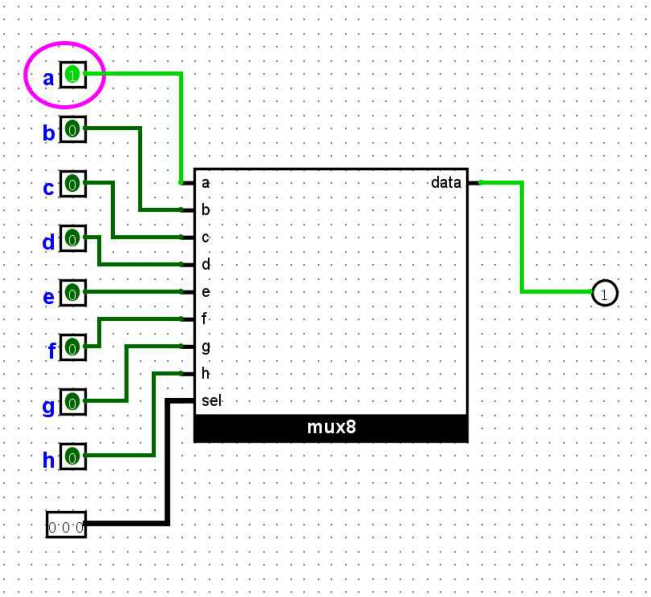
```
library ieee;
use ieee.std_logic_1164.all;

entity mux8 is
    port (
        a, b, c, d, e, f, g, h : in std_logic;
        sel : in std_logic_vector(2 downto 0);
        data : out std_logic
    );
end entity mux8;

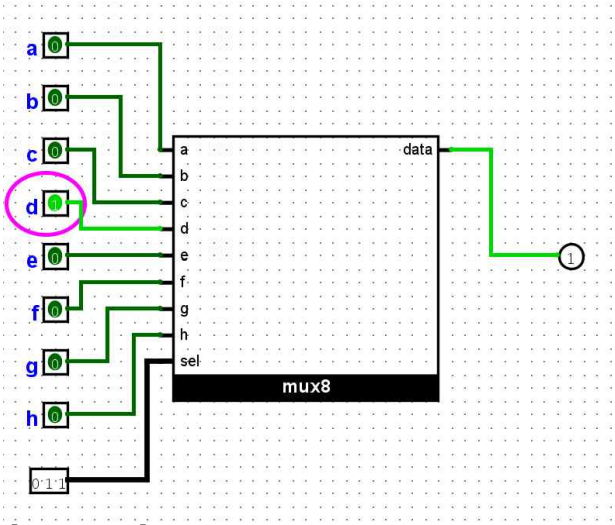
architecture bhv of mux8 is
begin
    process(a, b, c, d, e, f, g, h, sel)
```

```
    begin
        case sel is
            when "000" => data <= a;
            when "001" => data <= b;
            when "010" => data <= c;
            when "011" => data <= d;
            when "100" => data <= e;
            when "101" => data <= f;
            when "110" => data <= g;
            when "111" => data <= h;
            when others => data <= '0';
        end case;
    end process;
end architecture bhv;
```

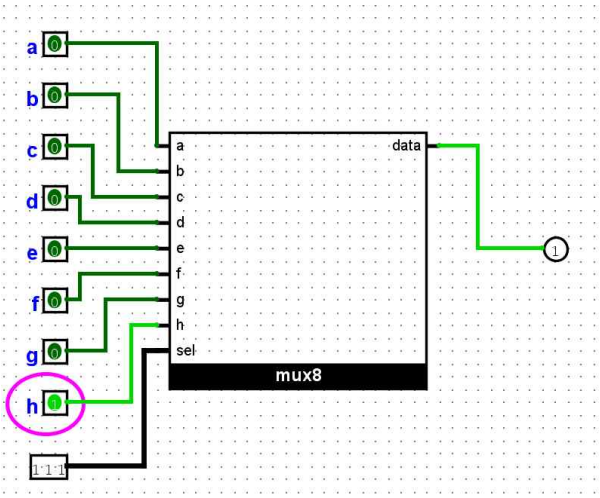
Simulation Screendshots



[picture 1]



[picture 2]



[picture 3]

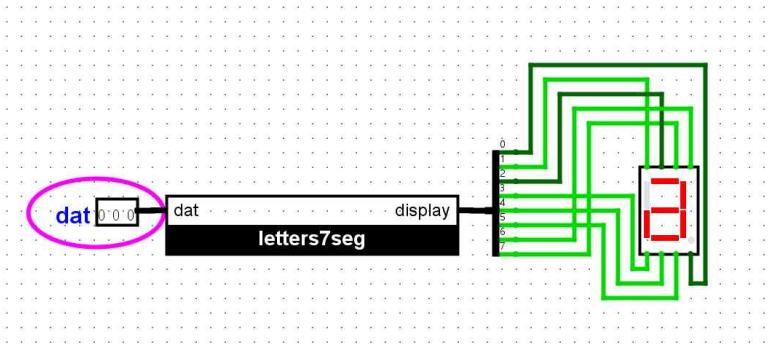
VHDL Code_3

```
library ieee;
use ieee.std_logic_1164.all;

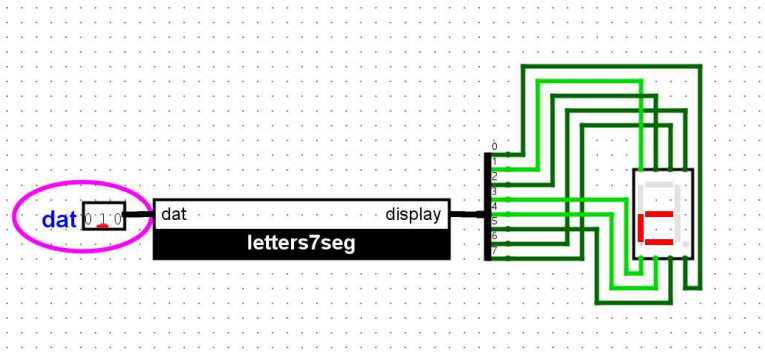
entity letters7seg is
    port (
        dat : in std_logic_vector(2 downto 0);
        display : out std_logic_vector(7 downto 0)
    );
end entity letters7seg;
```

```
architecture bhv of letters7seg is
begin
    process(dat)
    begin
        case dat is
            when "000" => display <= "11111010"; -- a
            when "001" => display <= "00111110"; -- b
            when "010" => display <= "00011010"; -- c
            when "011" => display <= "01111010"; -- d
            when "100" => display <= "10001110"; -- f
            when "101" => display <= "11110110"; -- g
            when others => display <= "00000000";
        end case;
    end process;
end architecture bhv;
```

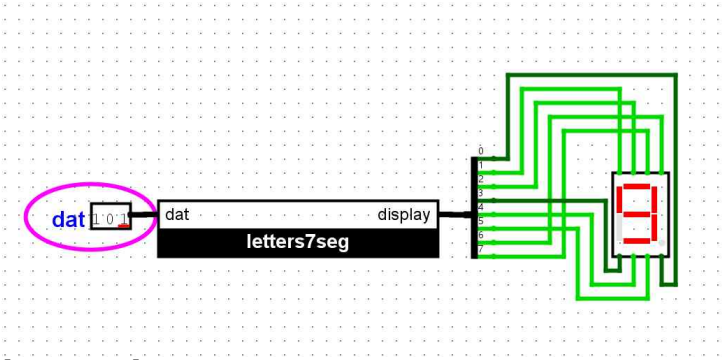
Simulation Screendshots



[picture 1]



[picture 2]



[picture 3]

VHDL Code_4

```
library ieee;
use ieee.std_logic_1164.all;

entity vote25 is
    port (
        vt : in std_logic_vector(24 downto 0);
        result : out std_logic
    );
end entity vote25;

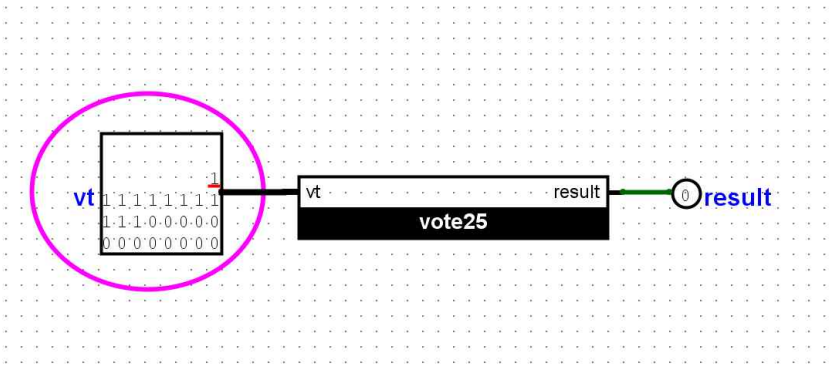
architecture bhv of vote25 is
begin
    process(vt)
        variable cnt : integer range 0 to 25;
    begin
        cnt := 0;

        for i in 0 to 24 loop
            if vt(i) = '1' then
                cnt := cnt + 1;
            end if;
        end loop;

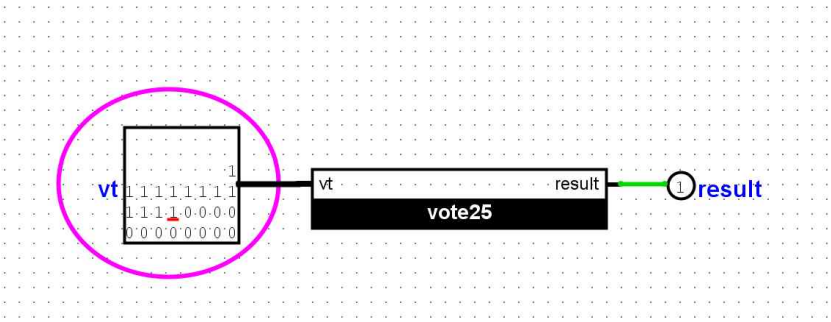
        if cnt >= 13 then
            result <= '1';
        else
            result <= '0';
```

```
        end if;
    end process;
end architecture bhv;
```

Simulation Screendshots



[picture 1]



[picture 2]

VHDL Code_5

```
library ieee;
use ieee.std_logic_1164.all;

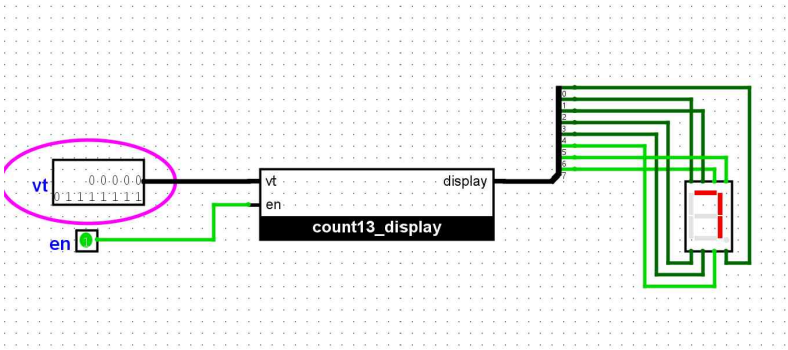
entity count13_display is
    port (
        vt : in std_logic_vector(12 downto 0);
        en : in std_logic;
        display : out std_logic_vector(7 downto 0)
    );
end entity count13_display;
```

```
architecture bhv of count13_display is
begin
    process(vt, en)
        variable cnt : integer range 0 to 15;
    begin
        cnt := 0;

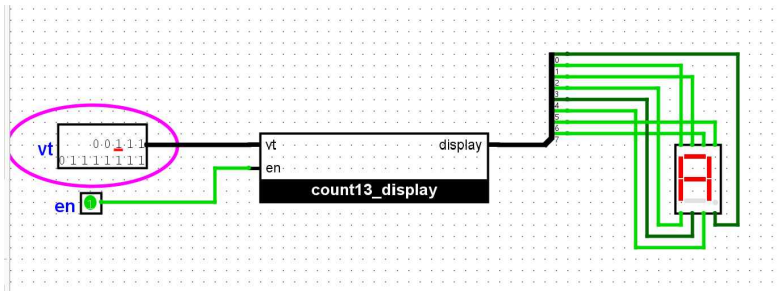
        for i in 0 to 12 loop
            if vt(i) = '1' then
                cnt := cnt + 1;
            end if;
        end loop;

        if en = '1' then
            case cnt is
                when 0  => display <= "11111100";
                when 1  => display <= "01100000";
                when 2  => display <= "11011010";
                when 3  => display <= "11110010";
                when 4  => display <= "01100110";
                when 5  => display <= "10110110";
                when 6  => display <= "10111110";
                when 7  => display <= "11100000";
                when 8  => display <= "11111110";
                when 9  => display <= "11110110";
                when 10 => display <= "11101110"; -- A
                when 11 => display <= "00111110"; -- b
                when 12 => display <= "10011100"; -- C
                when 13 => display <= "01111010"; -- d
                when 14 => display <= "10011110"; -- E
                when 15 => display <= "10001110"; -- F
                when others => display <= "00000000";
            end case;
        else
            display <= "00000000";
        end if;
    end process;
end architecture bhv;
```

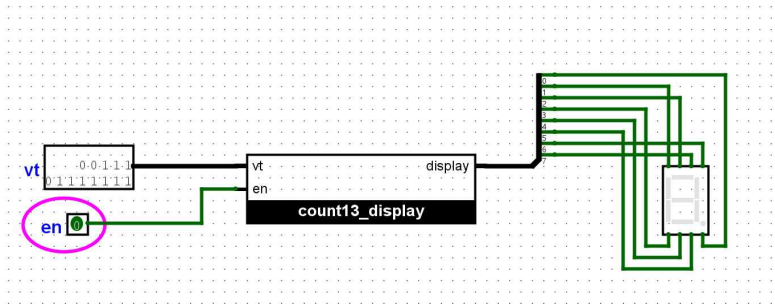
Simulation Screendshots



[picture 1]



[picture 2]



[picture 3]